

Internship Project — Week 1

Project Title: House Price Prediction

Assigned Date: [17/05/2026] **Submission Date:** [21/05/2026]

Problem Statement

Real estate buyers and sellers often rely on guesswork or outdated comparisons to estimate a property's fair value. Your task is to build a regression model that predicts house prices based on property features such as size, number of rooms, location, and age — and then identify which features most strongly influence price.

You will work with a real-world housing dataset, clean and prepare the data, train a regression model, evaluate its accuracy, and present your findings through charts and a short report.

Dataset — Where to Get It

Go to this link and download the dataset:

 <https://www.kaggle.com/datasets/yasserh/housing-prices-dataset>

Steps to download:

Create a free account on www.kaggle.com (if you don't have one)

Search for "Housing Prices Dataset"

Download the file — it will be a .csv file (usually named Housing.csv)

Use the full dataset as-is — it's small enough to work with directly

Tasks to Complete

Complete all the following tasks in a Jupyter Notebook (.ipynb):

Task 1 — Data Loading & Exploration

Load the CSV file using Pandas

Display the first 10 rows

Check how many rows and columns are there

Identify which column is the target (Price) and which are features

Check for missing values in each column

Task 2 — Data Cleaning

Handle missing values (fill or drop, whichever fits the column)

Remove duplicate rows, if any

Convert any categorical columns (e.g., yes/no fields) into numeric form using one-hot encoding

Keep only the columns that are meaningful for predicting price

Task 3 — Model Building

Split the data into training and test sets (80/20)

Train a Linear Regression model to predict house price

Evaluate it using MAE, RMSE, and R^2 Score

Train a second model — Random Forest Regressor — and compare its performance to the Linear Regression model

Task 4 — Visualization (Minimum 3 charts)

Chart 1: Histogram showing the distribution of house prices

Chart 2: Correlation heatmap showing which features relate most strongly to price

Chart 3: A chart of your own choice — be creative! (e.g., actual vs. predicted price scatter plot, price vs. area scatter plot, etc.)

Task 5 — Insights & Summary

Write a short paragraph (5–8 lines) inside your notebook answering:

Which features influence house price the most?

How accurate was your model (in plain terms)?

What surprised you in the data?

One recommendation for a real estate business based on your findings

Tools & Libraries to Use

Tool	Purpose
Python 3.x	Main programming language
Jupyter Notebook	To write and run your code
Pandas	Data loading and cleaning
Scikit-learn	Regression models and evaluation
Matplotlib / Seaborn	Charts and visualization

What to Submit

Create a folder named: **HousePricePrediction_[YourName]**

Inside the folder, include:

✓ analysis.ipynb — Your complete Jupyter Notebook with all 5 tasks

✓ Housing.csv — The dataset you used

✓ summary.pdf or summary.docx — A 1-page written summary of your findings (can be exported from the notebook)

✓ charts/ — A folder with all 3 chart images saved as .png

Submit the entire folder as a ZIP file

Via:

<https://docs.google.com/forms/d/e/1FAIpQLSf7t2YhRzjpQi0CL4TtG6gjqGWZ168Rk047SmuPTAXIUB7Flw/viewform?usp=dialog>